Done

String Array GIN Index

In this assignment, you will create a table of documents and then produce a GIN-based **text[]** reverse index for those documents that identifies each document which contains a particular word using SQL.

FYI: In *contrast* with the provided sample SQL, you will map all the words in the GIN index to *lower* case (i.e. Python, PYTHON, and python should all end up as "python" in the GIN index).

Connection Details:

Host: pg.pg4e.com

Port: 5432

Database: pg4e_402ae32925

User: pg4e_402ae32925

Password: (hide/show copy)

Check Answer

Here is the the **psql** command to access your database:

```
psql -h pg.pg4e.com -p 5432 -U pg4e_402ae32925 pg4e_402ae32925
```

The goal of this assignment is to run these queries:

```
SELECT id, doc FROM docs03 WHERE '{interpreter}' <@ string_to_array(lower(doc), ' ');
EXPLAIN SELECT id, doc FROM docs03 WHERE '{interpreter}' <@ string_to_array(lower(doc), ' ');
```

and (a) get the correct document(s) and (b) use the GIN index (i.e. not use a sequential scan).

```
CREATE TABLE docs03 (id SERIAL, doc TEXT, PRIMARY KEY(id));

CREATE INDEX array03 ON docs03 USING gin(...);
```

If you get an **operator class** error on your **CREATE INDEX** check the instructions below for the solution.

Here are the one-line documents that you are to insert into **docs03**:

```
INSERT INTO docs03 (doc) VALUES
('should be wondering a bit about the Python interpreter itself What'),
('language is it written in Is it written in a compiled language When we'),
('type python what exactly is happening'),
('The Python interpreter is written in a highlevel language called C'),
('You can look at the actual source code for the Python interpreter by'),
('source code So Python is a program itself and it is compiled into'),
('machine code When you installed Python on your computer or the vendor'),
('installed it you copied a machinecode copy of the translated Python'),
('program onto your system In Windows the executable machine code for'),
('Python itself is likely in a file with a name like');
```

Done

You should also insert a number of filler rows into the table to make sure PostgreSQL uses its index:

```
INSERT INTO docs03 (doc) SELECT 'Neon ' || generate_series(10000,20000);
```

It might take from a few seconds to a minute or two before PostgreSQL catches up with its indexing. The autograder won't work if the index is incomplete. If the **EXPLAIN** command says that it is using **Seq Scan -** the index has not completed yet. Run the above **EXPLAIN** multiple times if necessary until you verify that PostgreSQL has finished making the index:

```
pg4e=> EXPLAIN SELECT id, doc FROM docs03 WHERE '{interpreter}' <@ string_to_array(lower(doc), ' '
                                QUERY PLAN
-------------------------------------------------------------------------
 Seq Scan on docs03  (cost=0.00..177.24 rows=35 width=36)
 Filter: ('{interpreter}'::text[] <@ string_to_array(lower(doc), ' '::text))
(2 rows)


TIME PASSES......


pg4e=> EXPLAIN SELECT id, doc FROM docs03 WHERE '{interpreter}' <@ string_to_array(lower(doc), ' '
                                      QUERY PLAN
---------------------------------------------------------------------------------------
 Bitmap Heap Scan on docs03  (cost=12.02..21.97 rows=3 width=15)
 Recheck Cond: ('{interpreter}'::text[] <@ string_to_array(lower(doc), ' '::text))
   ->  Bitmap Index Scan on array03  (cost=0.00..12.02 rows=3 width=0)
   Index Cond: ('{interpreter}'::text[] <@ string_to_array(lower(doc), ' '::text))
(4 rows)

pg4e=>
```

References

Here are some links on monitoring the building of indexes:

- PostgreSQL Wiki: Index Maintenance (https://wiki.postgresql.org/wiki/Index_Maintenance)

- Stackoverflow: Monitoring Progress of Index Construction in PostgreSQL (https://dba.stackexchange.com/a/161992/206399)

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